

Reading

PREPARING TO READ

A BUILD VOCABULARY The words in **blue** are used in the reading passage. Read the text below. Then write the correct form of each word next to its definition.

In December 2015, representatives from almost every country got together in Paris, France, and agreed on one thing: that humanity's role in accelerating climate change had gotten out of hand, and that something **profound** needed to be done about it. In a rare moment of solidarity, these countries signed what the world refers to today as the Paris Agreement.

By doing so, each nation made a commitment to radically reduce or **eliminate** greenhouse gas emissions by **shifting** toward renewable energy. The goal was to collectively limit global temperature rise by 1.5 degrees Celsius by the end of the century. The agreement included several other terms: Participating nations had to create a transparent framework to ensure accountability and to allow clear targets and **criteria** to be set. In addition, larger nations had to support developing nations, which in general contribute much less to the problem.

While much progress has been made since the signing of the Paris Agreement, more needs to be done to ensure we don't experience the worst effects of climate change. The problem, though, is a challenging one: The same fuels that heat up our atmosphere power our economies and sustain our way of life. Nonetheless, if left unchecked, climate change will lead to extreme weather, widespread destruction, and perhaps even our **extinction**. Most experts **conclude** that drastic steps need to be taken now to ensure the **preservation** of our species.

1. _____ (n) standards by which something is assessed
2. _____ (n) the act of keeping something in its current state
3. _____ (adj) extreme, or deeply felt
4. _____ (v) to get rid of
5. _____ (n) the loss of an entire species
6. _____ (v) to arrive at an opinion
7. _____ (v) to move or change gradually



World leaders celebrate after the signing of the Paris Agreement.

B BUILD VOCABULARY Complete the sentences below with the correct form of the words in **blue**. Use a dictionary to help you.

erosion (n)
irony (n)

geology (n)
merit (n)

grasp (v)
perspective (n)

intriguing (adj)
subtle (adj)

1. The evidence he presented wasn't strong, so the committee didn't see much _____ in his argument.
2. It's a complex issue, and different people tend to have different _____ on it.
3. The two materials may seem the same at first glance, but if you study them carefully, you'll notice _____ differences in their texture.
4. Some people thought her idea was silly, but I found it _____.
5. The disaster was entirely their fault. The _____ is that they were brought in to prevent it.
6. Planting trees is a great way to reduce soil _____ caused by rain.
7. The concept is rather abstract and hard to _____.
8. She wants to study _____ because she's interested in the natural structures and materials that make up our planet.

C USE VOCABULARY Discuss these questions with a partner.

1. Do you think it's important to consider the **perspectives** of people who disagree with you? How does doing so benefit us?
2. Have you ever held a strong opinion, but **shifted** your position over time? What was the opinion, and why did it shift?
3. Have any of the ideas covered in this book had a **profound** impact on you? Which were they, and why did they affect you?
4. Can you think of examples that illustrate the concept of **irony**?

D PREDICT Look at the title of the reading passage and the chart on page 224. What do you think the Anthropocene refers to?

Critical Thinking

E BRAINSTORM Discuss the questions with a partner.

Critical Thinking

1. What do you think Earth was like before humans existed?
2. Which of the changes that have happened since then are because of humans?
3. What evidence or geological footprints will the humans of today leave behind for future humans?

Check your answers as you read the passage.

The Human Age

by Elizabeth Kolbert

The transformed
desert of Dubai, U.A.E.

Human beings have altered the planet so much in just the past century or two that we are now in a new epoch: the Anthropocene.

A  The word Anthropocene was coined by Dutch chemist Paul Crutzen in 2002. Crutzen, who shared a Nobel Prize for discovering the effects of ozone-depleting compounds, was sitting at a scientific conference one day. The conference chairman kept referring to the Holocene, the epoch that began 11,500 years ago, at the end of the last ice age, and that—officially, at least—continues to this day.

B “Let’s stop it,” Crutzen recalls blurting out. “We are no longer in the Holocene. We are in the Anthropocene.” It was quiet in the room for a while. When the group took a coffee break, the Anthropocene was the main topic of conversation.

C Way back in the 1870s, an Italian geologist named Antonio Stoppani proposed that people had introduced a new era, which he labeled the Anthropozoic. Stoppani’s proposal was ignored; other scientists found it unscientific. The Anthropocene, by contrast, struck a chord. The human impact on the world had become a lot more obvious since Stoppani’s day, in part because the size of the population had roughly quadrupled,¹ to nearly seven billion.

D When Crutzen wrote up the Anthropocene idea in the journal *Nature*, the concept was immediately picked up by researchers working in a wide range of disciplines. Soon, it began to appear regularly in the scientific press. At first, most of the scientists using the new geologic term were not geologists. Jan

Zalasiewicz, a British geologist, found the discussions **intriguing**. “I noticed that Crutzen’s term was appearing in the serious literature, without quotation marks and without a sense of **irony**,” he says.

E In 2007, Zalasiewicz was serving as chairman of the Geological Society of London’s Stratigraphy² Commission. At a meeting, he decided to ask his fellow stratigraphers what they thought of the Anthropocene. Twenty-one of twenty-two thought the concept had **merit**. The group agreed to look at it as a formal problem in **geology**. Would the Anthropocene satisfy the **criteria** used for naming a new epoch?

F The rock record of the present doesn’t exist yet, of course. So the question was: When it does, will human impacts show up as “stratigraphically significant”? The answer, Zalasiewicz’s group decided, is yes—though not necessarily for the reasons you would expect.

G Probably the most obvious way humans are altering the planet is by building cities, which are essentially vast stretches of man-made materials—steel, glass, concrete, and brick. But it turns out most cities are not good candidates for long-term **preservation**: they’re built on land, and on land the forces of erosion tend to win out over those of sedimentation. From a geologic **perspective**, the most plainly visible human effects on the landscape today “may in some ways be the most transient,³” Zalasiewicz observes.

¹ If something **quadruples**, it increases by a factor of four.

² **Stratigraphy** is a branch of geology concerned with the study of rock layers.

³ **Transient** describes a situation that lasts only a short time or is constantly changing.

Earth's Geological Timeline

Start of the
Anthropocene

Era	Period		Epoch	Millions of Years Ago
Cenozoic	Quaternary		Holocene	
			Pleistocene	
	Neogene		Pliocene	
			Miocene	
			Oligocene	
	Paleogene		Eocene	
			Paleocene	
Mesozoic	Cretaceous			
	Jurassic			
	Triassic			
Paleozoic	Permian			
	Carboniferous	Pennsylvanian		
		Mississippian		
	Devonian			
	Silurian			
	Ordovician			
	Cambrian			
Precambrian	Proterozoic			540
	Archean			2500
				3800
	Hadean			4600

In geology, *epochs* are relatively short time spans, though they can extend for tens of millions of years. *Periods*, such as the Ordovician and the Cretaceous, last much longer, and *eras*, like the Mesozoic, longer still. The boundaries between epochs are defined by changes preserved in sedimentary rocks⁴—for example, the emergence of one type of commonly fossilized organism, or the disappearance of another.

H Humans have also transformed the world through farming; more than half of the planet's habitable land is now devoted to agriculture. Here again, some of the effects that seem most significant today—runoff from the use of fertilizers on fields, for example—will leave behind only **subtle** traces at best. Future geologists are most likely to **grasp** the scale of 21st-century industrial

agriculture from the pollen⁵ record—from the monochrome⁶ stretches of corn, wheat, and soy pollen that will have replaced the varied record left behind by rainforests or prairies.

I The leveling of the world's forests will send at least two coded signals to future stratigraphers, though deciphering the first may be tricky. Massive soil **erosion** is causing increasing sedimentation⁷ in

⁴ Sedimentary rocks are formed from sediment—solid material that settles at the bottom of a liquid.

⁵ Pollen is a powder produced by flowers that fertilizes other flowers of the same species.

⁶ If something is **monochrome**, it has just one color.

⁷ Sedimentation is the process by which solid material—especially earth and pieces of rock—settles at the bottom of a liquid.

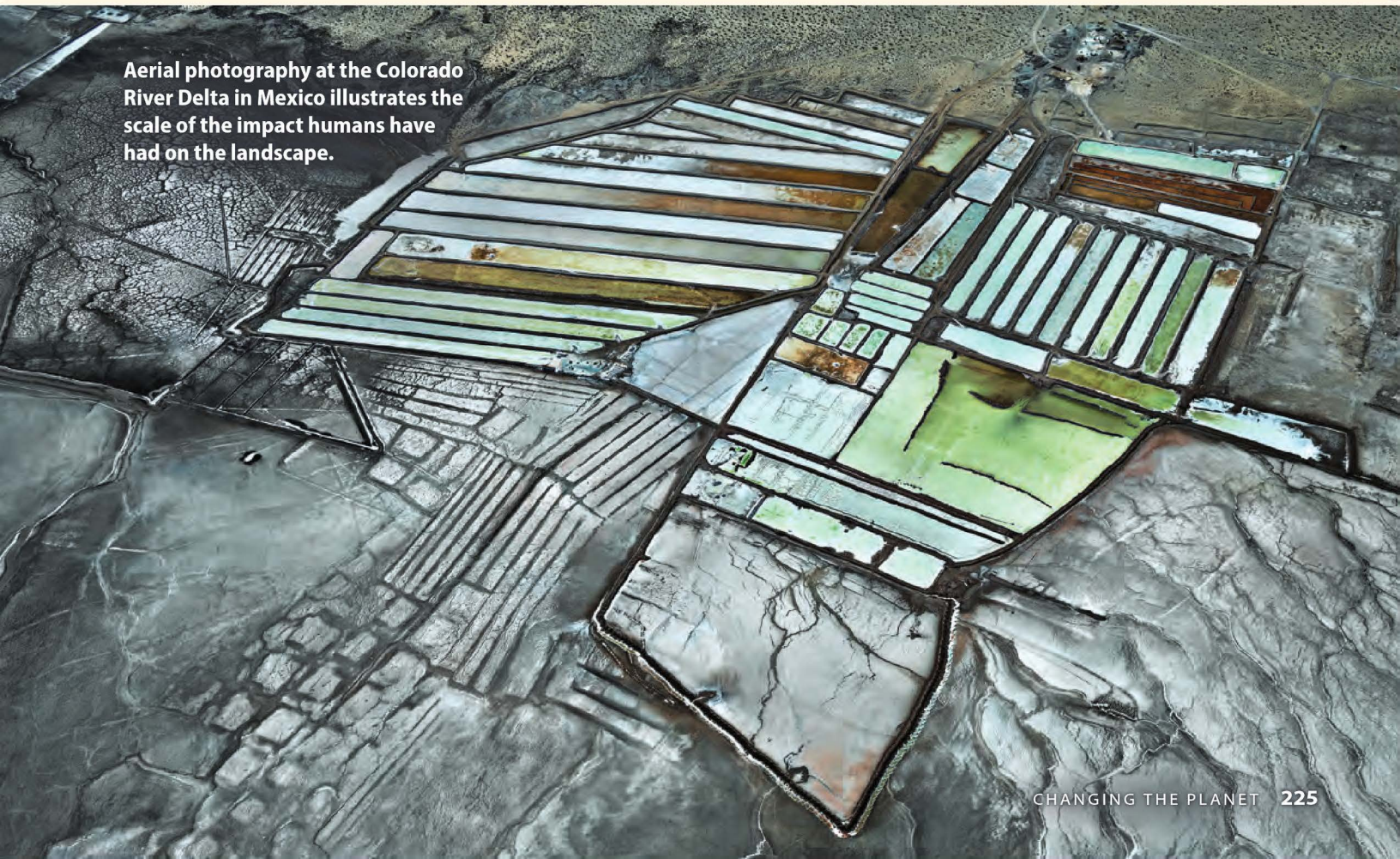
some parts of the world—but at the same time, the dams we’ve built on most of the world’s major rivers are holding back sediment that would otherwise be washed to sea. The second signal of deforestation should come through clearer. Loss of forest habitat is a major cause of **extinctions**, which are now happening at a rate hundreds or even thousands of times higher than during most of the past half billion years. If current trends continue, the rate may soon be tens of thousands of times higher.

J Probably the most significant change, from a geologic perspective, is one that’s invisible to us—the change in the composition of the atmosphere. Carbon dioxide emissions are colorless, odorless, and—in an immediate sense—harmless. But their warming effects could easily push global temperatures to levels that have not been seen for millions of years. Some plants and animals are already

shifting their ranges toward the Poles, and those shifts will leave traces in the fossil record. Some species will not survive the warming at all. Meanwhile, rising temperatures could eventually raise sea levels 20 feet (6 meters) or more.

K Long after our cars, cities, and factories have turned to dust, the consequences of burning billions of tons’ worth of coal and oil are likely to be clearly discernible. As carbon dioxide warms the planet, it also seeps into the oceans and acidifies them. Sometime this century, they may become acidified to the point that corals can no longer construct reefs, which would register in the geologic record as a “reef gap.” Reef gaps have marked each of the past five major mass extinctions. The most recent one—which is believed to have been caused by the impact of an asteroid—took place 65 million years ago, at the end of the Cretaceous period; it **eliminated** not just

Aerial photography at the Colorado River Delta in Mexico illustrates the scale of the impact humans have had on the landscape.



the dinosaurs but also the plesiosaurs, pterosaurs, and ammonites.⁸ Since then, there has been nothing to match the scale of the changes that we are now seeing in our oceans. To future geologists, Zalasiewicz says, our impact may look as sudden and **profound** as that of an asteroid.

- L** If we have indeed entered a new epoch, then when exactly did it begin? When did human impacts rise to the level of geologic significance?

M William Ruddiman, a paleoclimatologist at the University of Virginia, proposed that the invention of agriculture some 8,000 years ago—and the deforestation that resulted—led to an increase in atmospheric CO₂ just large enough to stave off what otherwise would have been the start of a new ice age. In his view, humans have been the dominant force on the planet practically since the start of the Holocene. Crutzen suggested that the Anthropocene began

⁸ Plesiosaurs, pterosaurs, and ammonites are extinct prehistoric animal species.

Rocks structures like these in Isle of Skye, Scotland, contain records of millions of years of history.



in the late 18th century, when, ice cores show, carbon dioxide levels began what has since proved to be an uninterrupted rise. Other scientists put the beginning of the new epoch in the middle of the 20th century, when the rates of both population growth and consumption accelerated rapidly.

N To answer the question definitively, a working group of the International Commission on Stratigraphy (ICS) was assembled. Initially headed by Zalasiewicz, the group was tasked

with officially determining whether the Anthropocene deserves to be incorporated into the geologic timescale. As expected, the investigation dragged on for many years. However, the decision did become easier over time as the impact of humans on the planet grew and became even more stratigraphically significant. In 2016, a decision was finally reached. The working group agreed that the Anthropocene is indeed a distinct epoch, separate from the Holocene. They also **concluded** that the human age began in the year 1950, when the Great Acceleration—a dramatic increase in the rate of population growth and human activity affecting the planet—took off.

- o For Crutzen, who started the debate, the real value of recognizing the Anthropocene doesn't lie in the revisions that have to be made to geology textbooks. His purpose is broader: He wants to focus our attention on the consequences of our collective action—and on how we might still avert the worst. "What I hope," he says, "is that the term Anthropocene will be a warning to the world."

Adapted from "The Age of Man," by Elizabeth Kolbert: National Geographic Magazine, March 2011

Elizabeth Kolbert is an award-winning writer who has written extensively about environmental issues for *National Geographic Magazine*, *The New Yorker*, and other publications. Her book *The Sixth Extinction* won the 2015 Pulitzer Prize for general nonfiction.



UNDERSTANDING THE READING

A UNDERSTAND THE MAIN IDEA What does “Anthropocene” mean? Explain it in your own words.

B UNDERSTAND MAIN IDEAS Skim paragraphs G–K. What are the four main areas of human impact the author covers?

1. cities 3. _____
2. _____ 4. _____

Critical Thinking

C SUMMARIZE Complete the chart summarizing the human impact on our planet.

Four areas of human impact	How it will change the planet	Details
1. <i>cities</i>		
2.		
3.		
4.		

Critical Thinking

D INTERPRET VISUAL DATA Look at the timeline on page 224 and note answers to the questions below. Then discuss your ideas with a partner.

1. What era, period, and epoch are we currently living in?
- Era: _____ Period: _____
- Epoch: _____
2. When did the current era begin? _____
3. How do scientists decide when one epoch ends and another one begins?

Review
this Critical
Thinking Skill
in Unit 8